

Project 4b: Agent-Based Modeling with NetLogo

A Spatially Explicit Model of Wildfire Spread and Suppression

The 2021 Marshall Fire, Boulder County, Colorado

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1. Introduction

Agent-based modeling (ABM) is a computational approach that simulates the interactions of individual entities within an environment in order to understand how complex system-level patterns emerge from localized rules. In contrast to traditional deterministic or equation-based models, ABMs allow for spatial heterogeneity, stochastic processes, and dynamic interactions between agents and their environment. These characteristics make ABMs particularly suitable for modeling natural hazards such as wildfires, where local conditions, randomness, and environmental variability play critical roles in determining overall behavior (Crooks et al. 2018, 3).

This project develops a spatially explicit wildfire ABM using NetLogo, integrating real-world geographic data to simulate the spread and suppression of the 2021 Marshall Fire in Boulder County, Colorado. The Marshall Fire ignited on December 30, 2021, near the Marshall area south of Boulder and spread rapidly eastward into the communities of Superior and Louisville under extreme wind conditions. It destroyed more than one thousand homes and became one of the most destructive urban-interface wildfires in Colorado history (Natural Hazards Center 2022, 1). The fire is particularly relevant for spatial modeling because its behavior was strongly influenced by land cover, wind direction, transportation infrastructure, and human response.

The goal of this project is not to reproduce the Marshall Fire with full physical accuracy, but to construct a spatially grounded and interpretable model that demonstrates how wildfire behavior emerges from interactions between fuel conditions, environmental forcing, and suppression efforts. By linking GIS data with NetLogo, the model explores how spatial structure influences both fire spread and response effectiveness.

2. Spatial Data

Three spatial datasets were used in this project. Each was selected because it provided information essential to making the model meaningfully spatial rather than abstract.

2.1 NLCD 2021 Land Cover Raster

The primary input was the NLCD 2021 land cover raster obtained from the Multi-Resolution Land Characteristics Consortium (mrlc.gov). This raster provided a classified land cover surface covering the Marshall Fire study area. Each cell carries an integer code identifying the land cover type, including open water, developed land at four intensity levels, barren land, deciduous and evergreen forest, shrub and scrub, grassland, pasture, cultivated crops, and wetlands. This layer was essential because fire behavior depends fundamentally on fuel type. Without a spatially explicit fuel surface, every patch in the model would be equally flammable and the simulation would produce no meaningful geographic pattern (Homer et al. 2020, 4). The raster was processed in ArcGIS Pro, reprojected to WGS84, clipped to the study area, resampled to a coarser cell size for computational manageability, and exported as an ESRI ASCII grid for import into NetLogo.

2.2 Roads Shapefile

The second input was a roads shapefile derived from the Census TIGER 2023 dataset for Boulder County, clipped and reprojected to match the study area. Roads served two distinct roles in the model. First, they functioned as partial barriers to fire spread, with road patches receiving a flammability reduction controlled by the road-crossing-factor slider. Second, they served as the preferred travel surface for firefighters before

they engaged the fire line, giving the firefighter movement rules a realistic geographic constraint rooted in the actual street network of Superior and Louisville rather than straight-line Euclidean movement.

2.3 Marshall Fire Burn Perimeter

The third input was the Marshall Fire burn perimeter obtained from the National Interagency Fire Center WFIGS 2021 wildfire perimeters dataset. This layer was not imported into NetLogo directly. Instead it served two purposes in ArcGIS Pro: defining the geographic extent of the study area and serving as a reference boundary for comparing the simulated burn output against the observed fire footprint. The study area was defined to extend somewhat beyond the documented burn perimeter to give the model room to explore spread under different parameter settings without restricting the simulation to only the already-burned zone.

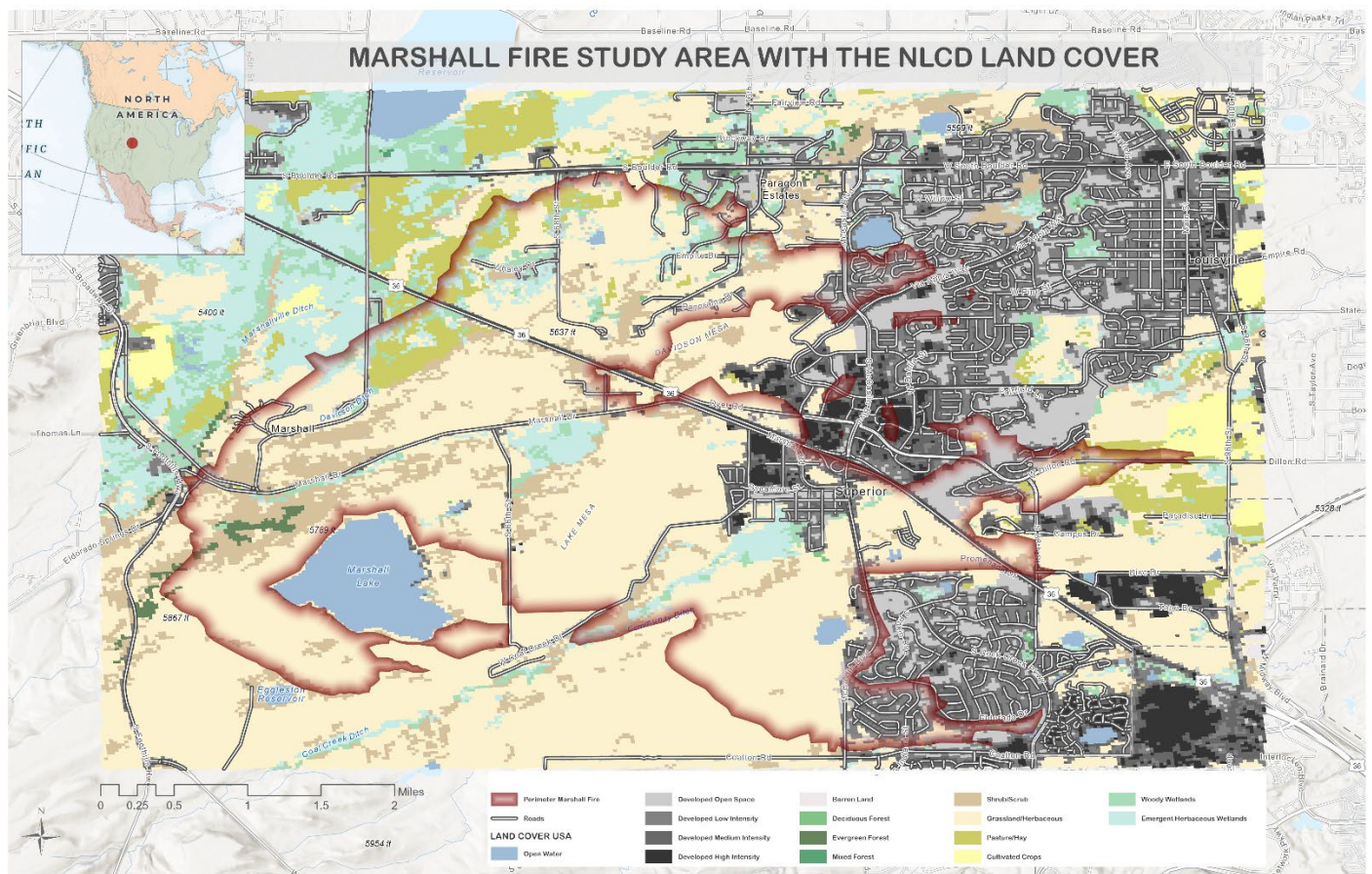


Figure 1. Study area map showing NLCD 2021 land cover, Marshall Fire perimeter, and road network, Boulder County, Colorado.

3. Methods

3.1 Starting Point and Overall Strategy

The model was built by editing the standard NetLogo Fire model from the Models Library. The original model randomly distributed tree patches at a user-specified density, ignited the entire left edge of the world, and spread fire to adjacent green patches while producing a fading ember visual effect. It had no connection to real spatial data, no differentiation in patch flammability, and only one agent type. The strategy for this project was to replace all of those elements progressively: first link the model to real GIS data, then assign spatially differentiated flammability, then add wind and spotting dynamics, and finally introduce firefighters as a second breed with station-based dispatch and adaptive movement rules. This progressive approach mirrors the workflow Crooks et al. recommend for linking an existing ABM to real geographic data (Crooks et al. 2018, 121).

3.2 Data Preparation in ArcGIS Pro

Before any coding could begin, the spatial data had to be prepared in a form that NetLogo's GIS extension could accept. NetLogo's GIS extension supports ESRI ASCII grid files for rasters and ESRI shapefiles for vector data, and it requires that raster files be accompanied by a valid projection file in a recognized format (Northwestern University Center for Connected Learning 2024).

The land cover raster presented the first major challenge: several NLCD annual products were loaded initially, but one of the annual layers showed incorrect or incomplete coverage for the study area. After testing multiple options, the legacy land cover layer proved most reliable. It was reprojected from its native NAD83 Albers coordinate system to WGS84 geographic coordinates, clipped to the study area extent, resampled to a coarser cell size to make the NetLogo world computationally manageable, and exported as an ESRI ASCII grid. The roads shapefile was similarly reprojected to WGS84 and clipped before being copied to the NetLogo project folder.

A resampling step was necessary because very fine rasters create extremely large NetLogo worlds with thousands of patches that slow execution significantly and make interactive debugging impractical. Each patch in the final model represented a meaningful ground area while keeping the total number of patches within a range that allowed the model to run interactively. This is the kind of trade-off between spatial resolution and computational feasibility that Crooks et al. describe as a central consideration in building spatially explicit ABMs (Crooks et al. 2018, 108). Rather than converting roads to raster at the same cell size, which would make them unrealistically wide and too strong as barriers, the roads were retained as a vector shapefile and loaded directly into NetLogo. This preserved their geometry better for intersection tests and allowed more flexible firefighter movement logic.

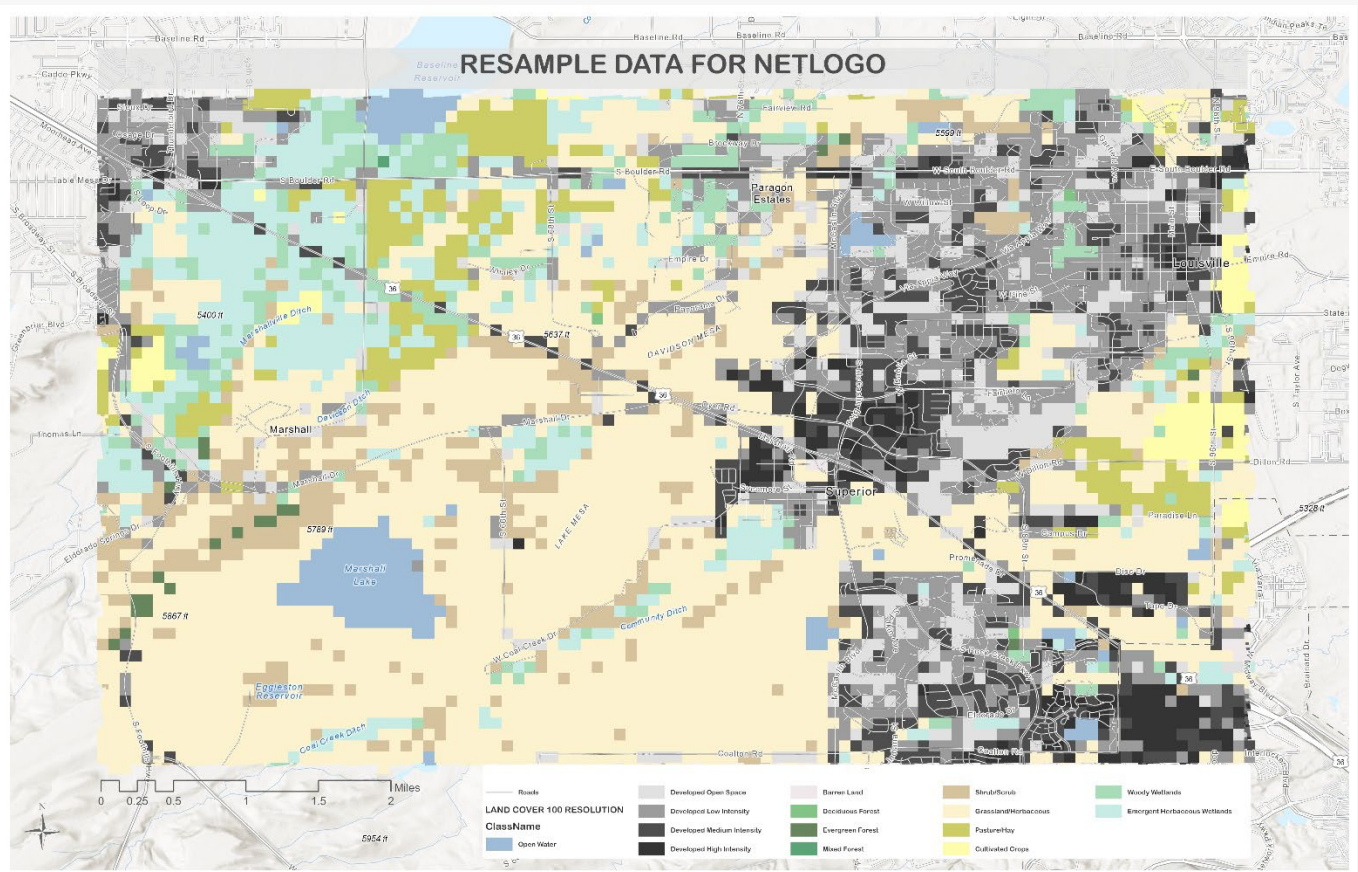


Figure 2. ArcGIS Pro project showing clipped NLCD land cover raster after being resampled 100 resolution.

3.3 Loading Spatial Data into NetLogo

The two key commands for importing data were `gis:load-dataset` and `gis:apply-raster`. The land cover raster was loaded into a global variable called `nlcd-dataset` and the roads shapefile was loaded into `roads-dataset`. The world envelope was set to match the raster extent using `gis:set-world-envelope` so that the NetLogo coordinate system aligned with the GIS data. The raster values were then transferred to patches using `gis:apply-raster nlcd-dataset lc`, which populated the `lc` patch variable with the NLCD integer code for each cell. Roads were identified using `gis:intersects?`, which tested whether each patch geometrically intersected any road feature in the shapefile and assigned a Boolean `road?` attribute accordingly (Northwestern University Center for Connected Learning 2024).

Before loading any data, the model established a coordinate system using `gis:set-coordinate-system` with a WGS84 well-known text string. This ensured all data shared the same geographic reference frame and allowed the `gis:project-lat-lon` primitive to correctly convert the documented fire origin coordinates from latitude and longitude into NetLogo patch coordinates for georeferenced ignition.

3.4 Flammability Assignment

Once each patch had an NLCD code, a reporter function called `flammability-from-lc` translated each code into a numeric flammability value between 0 and 1. These values were not derived from empirical fire behavior models but were designed to encode a defensible relative ordering of combustibility across land cover types. Open water was assigned zero. Developed land classes received very low values ranging from 0.04 for high-intensity development to 0.12 for open space. Forest classes received values between 0.78 and 0.88. Shrub and grassland were assigned values of 0.70 and 0.68 respectively, reflecting the high fuel availability in the dry Front Range grasslands that National Weather Service documentation identifies as a primary driver of the Marshall Fire's rapid spread (National Weather Service Boulder 2021). Road patches received a flammability reduction controlled by the `road-crossing-factor` slider rather than an absolute zero, reflecting that roads do not become absolute firebreaks under extreme ember transport and wind-driven spread.

NLCD Code	Land Cover Type	Flammability	Category
11	Open Water	0.00	Non-burnable
21	Developed Open Space	0.12	Very Low
22	Developed Low Intensity	0.10	Very Low
23	Developed Medium Intensity	0.07	Very Low
24	Developed High Intensity	0.04	Very Low
31	Barren Land	0.12	Low
41	Deciduous Forest	0.78	High
42	Evergreen Forest	0.88	Very High
43	Mixed Forest	0.82	High
52	Shrub/Scrub	0.70	High
71	Grassland/Herbaceous	0.68	High
81	Pasture/Hay	0.48	Moderate
82	Cultivated Crops	0.38	Moderate
90	Woody Wetlands	0.22	Low
95	Emergent Herbaceous Wetlands	0.12	Very Low

Table 1. NLCD land cover codes and assigned flammability values used in the model.

3.5 Wind and Directional Spread

A symmetric fire spread model could not produce a convincing Marshall Fire simulation because the documented event was driven overwhelmingly by strong westerly mountain-wave winds. Wind was incorporated by adding directional bias to the spread probability calculation inside the spread-fire procedure. Each potential ignition event calculated a directional bonus based on the alignment between the spread direction and the wind vector. Patches directly downwind received the full wind-strength bonus. Partially aligned neighbors received a reduced bonus at 55 percent of wind-strength. The baseline spread term was kept deliberately low at 0.02 so that wind and flammability structure dominated the spread pattern rather than allowing fire to spread aggressively in all directions regardless of orientation. A spotting

mechanism was added through the spotting-prob slider, allowing burning patches to probabilistically ignite cells several patches ahead of the immediate fire front in the downwind direction. This represented the wind-thrown ember transport that is documented as a primary mechanism in suburban wildfire environments (Natural Hazards Center 2022, 8).

3.6 Ignition Point

The default left-edge ignition of the original model was replaced with a georeferenced cluster centered on an approximate origin location near the Marshall area south of Boulder, using a latitude of 39.952709 and a longitude of -105.231. The `gis:project-lat-lon` primitive converted these coordinates into NetLogo patch space under the current GIS projection assumptions. A small cluster of patches within radius two around the center point was ignited rather than a single cell, because single-cell ignition was sometimes too brittle on a coarse patch grid where one cell might happen to be poorly representative of the surrounding fuel.

3.7 Firefighter Agents

Firefighters were added as a third breed to satisfy the requirement for at least two distinct agent types and to give the model a meaningful second dynamic beyond fire spread. Each firefighter was initialized with an energy value, an engagement status flag, a squad identifier, and tracking variables for detecting stuck behavior. Two fixed station locations were computed at setup: one in the upper-right portion of the study area representing Louisville Station 1, and a second on the right-middle side to represent an additional eastern response location. Splitting deployment between two stations produced more realistic approach geometry than concentrating all crews at a single point.

Firefighter movement used a two-phase logic. Before engaging the fire, crews traveled preferentially along road patches and through traversable land cover classes including grassland, pasture, developed open space, and barren land. After coming within the engagement radius of an active fire or ember agent, the firefighter switched to free-movement mode and advanced directly toward the fire line. If a firefighter stopped making progress over several consecutive ticks, a stuck-recovery rule advanced the agent directly toward its target and forced engagement, preventing permanent immobilization in difficult local geometry. Firefighters were organized into squads of five using a squad-id derived from the agent's who number, with different squads assigned to different sectors of the active fire to diversify approach routes and improve spatial coverage. This squad structure addresses the emergent problem of agent congestion that Crooks et al. describe as a common challenge in multi-agent movement models (Crooks et al. 2018, 87).

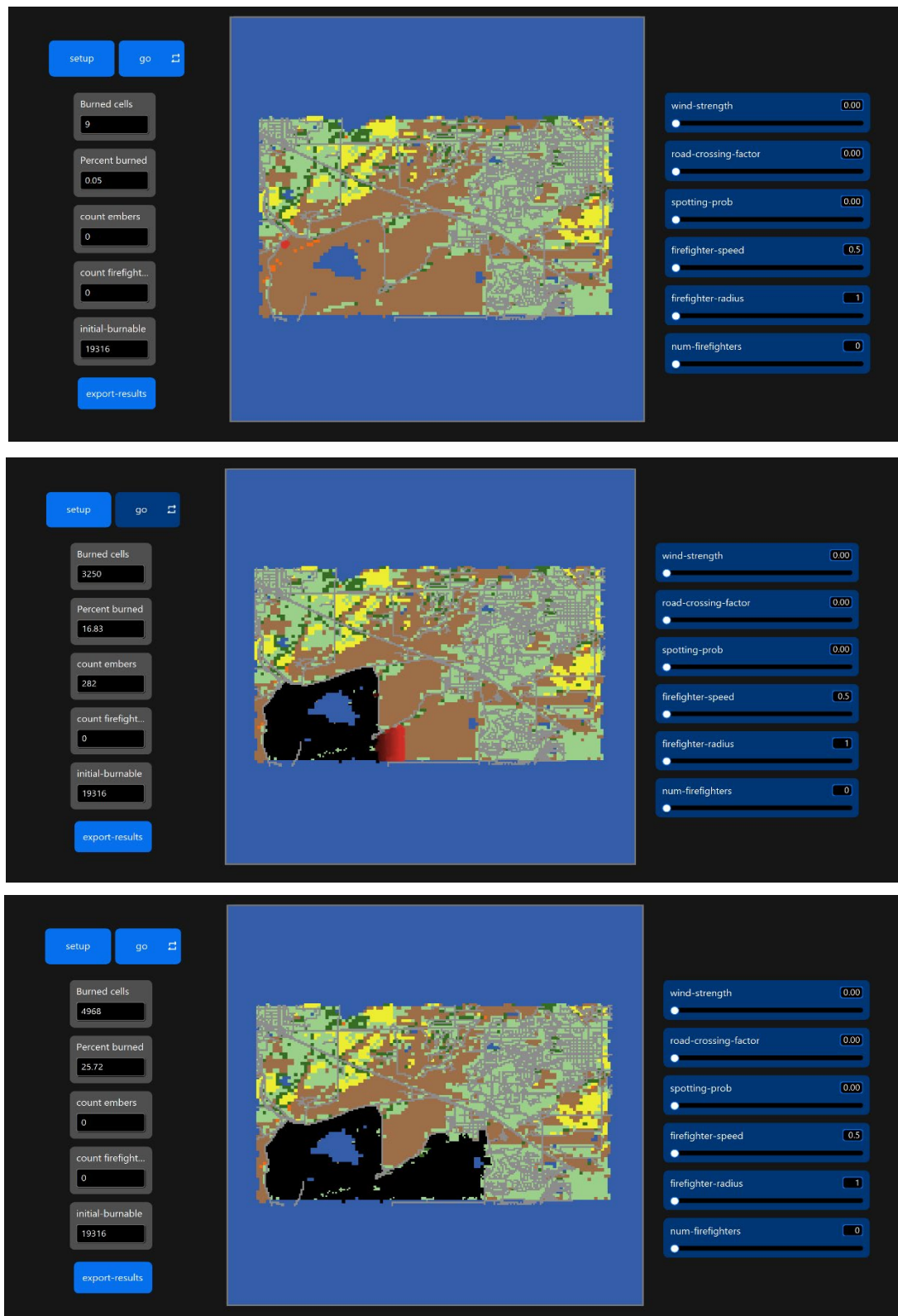


Figure 3. NetLogo interface -- after Setup, before Go NetLogo model world after Setup procedure completes. Landscape is colored by flammability class. Firefighter agents are yet not visible, just fire moving in min conditions.

3.8 Model Parameters

Many of the parameter settings were determined through repeated testing rather than arbitrary preference. Road-crossing-factor was kept low because even small nonzero values made the fire excessively destructive given the density of the road network in the suburban study area. Wind direction was set to an eastward vector matching the documented westerly-to-easterly forcing of the event. Spotting probability was set at a modest value to allow the fire to bridge barriers episodically without spreading so aggressively that suppression became irrelevant. Firefighter speed was set high enough that crews could meaningfully affect the fire before the simulation was effectively over, which was necessary to allow the second agent type to interact with the fire in an observable way. This parameter-tuning process is consistent with the calibration and sensitivity testing approach described by Crooks et al. (2018, 134).

3.9 Export and Return to ArcGIS Pro

At the end of each run, an export procedure saved two raster outputs using `gis:patch-dataset` and `gis:store-dataset`. The first output, `final_output.asc`, recorded the final state of every patch as a categorical value: 0 for nonburnable, 1 for unburned burnable land, 2 for burned, and 3 for roads unless burned. The second output, `burn_tick.asc`, recorded the simulation tick at which each patch first ignited, with unburned patches retaining a value of negative one. These files were brought back into ArcGIS Pro for final cartographic presentation and comparison with the observed Marshall Fire perimeter. This separation of simulation and final spatial presentation follows the workflow described by Crooks et al. in which NetLogo handles agent-based processing and GIS software handles polished output mapping (Crooks et al. 2018, 145).

The burn-tick variable required moving reset-ticks before the ignition cluster procedure at setup so that the ticks variable existed when the first patches ignited. Classification in `update-final-state` was ordered deliberately: nonburnable patches received 0, burnable patches 1, road patches 3, and burned patches 2 last, so that if a road cell burned it would be classified as burned rather than retaining the road classification.

4. Results

4.1 Landscape Visualization

The model successfully reproduces the dominant eastward spread pattern of the Marshall Fire. Fire spreads rapidly through grassland areas and slows when encountering developed areas and infrastructure. The overall shape of the simulated burn area shows strong directional bias consistent with wind-driven spread. After setup, the model world displayed a spatially differentiated landscape matching the land cover of the Marshall Fire study area. Water features appeared blue. Road patches appeared gray. Lower-flammability patches appeared in olive and brown tones. Higher-flammability patches in the foothill and grassland areas appeared in brighter greens. The two firefighter stations were visible as clusters of white agents in the eastern portion of the study area. The distribution of flammability classes was clearly visible and reflected the actual land cover gradient from forested foothills in the west through grassland and suburban development in the east.

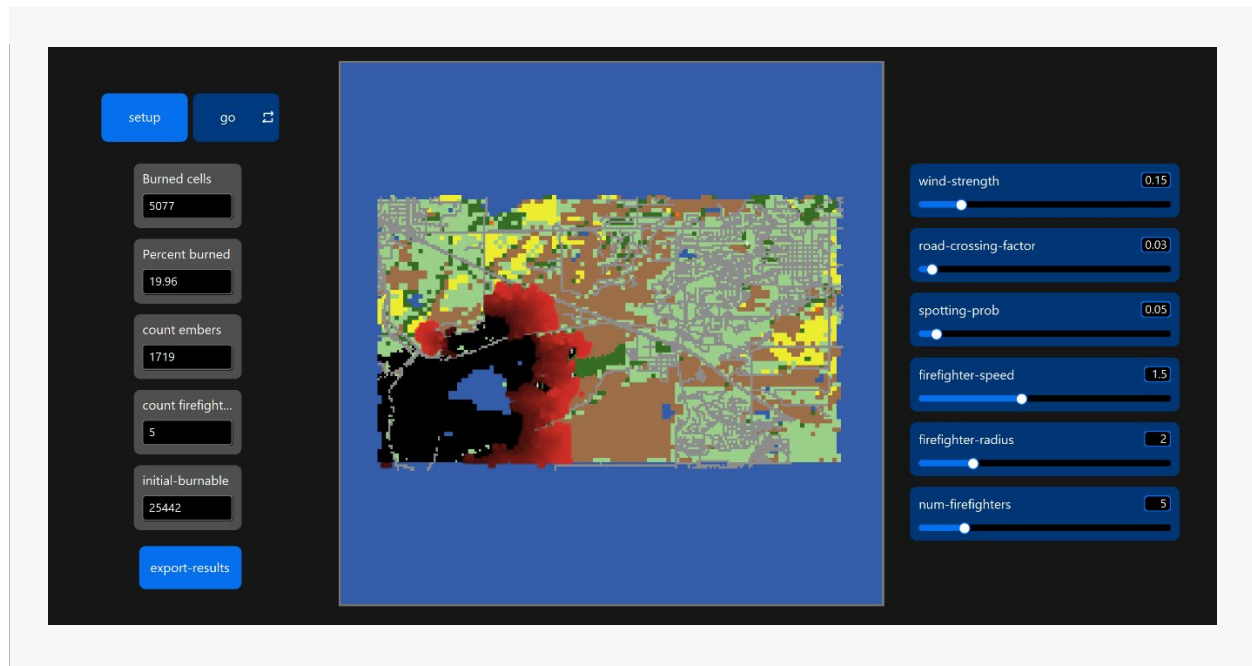


Figure 4. NetLogo model window during a mid-run state showing directional fire spread eastward from the origin cluster. Wind-driven spread pattern is visible. Firefighter agents are moving toward the fire line from station locations.

4.2 Fire Spread Pattern

With wind-strength set to a moderate value and road-crossing-factor kept low, the fire spread from the origin cluster predominantly eastward, consistent with the documented Marshall Fire spread direction. The grassland and shrubland patches in the western and central portions of the study area supported rapid advance while developed patches in the eastern portion showed slower and more fragmented ignition. Road patches introduced visible channeling effects that slowed spread in some corridors while spotting allowed the fire to bridge gaps in others. The heterogeneous burn pattern, with some developed patches remaining unburned while others burned due to ember deposition, visually matched the patchy structure documented in post-fire damage assessments (Natural Hazards Center 2022, 12).

Firefighters reduce fire spread locally, but their effectiveness is limited when the fire expands rapidly. In scenarios with moderate wind and lower spotting probability, suppression has a noticeable impact. However, under stronger wind conditions, the fire spreads faster than firefighters can respond, reducing their effectiveness.

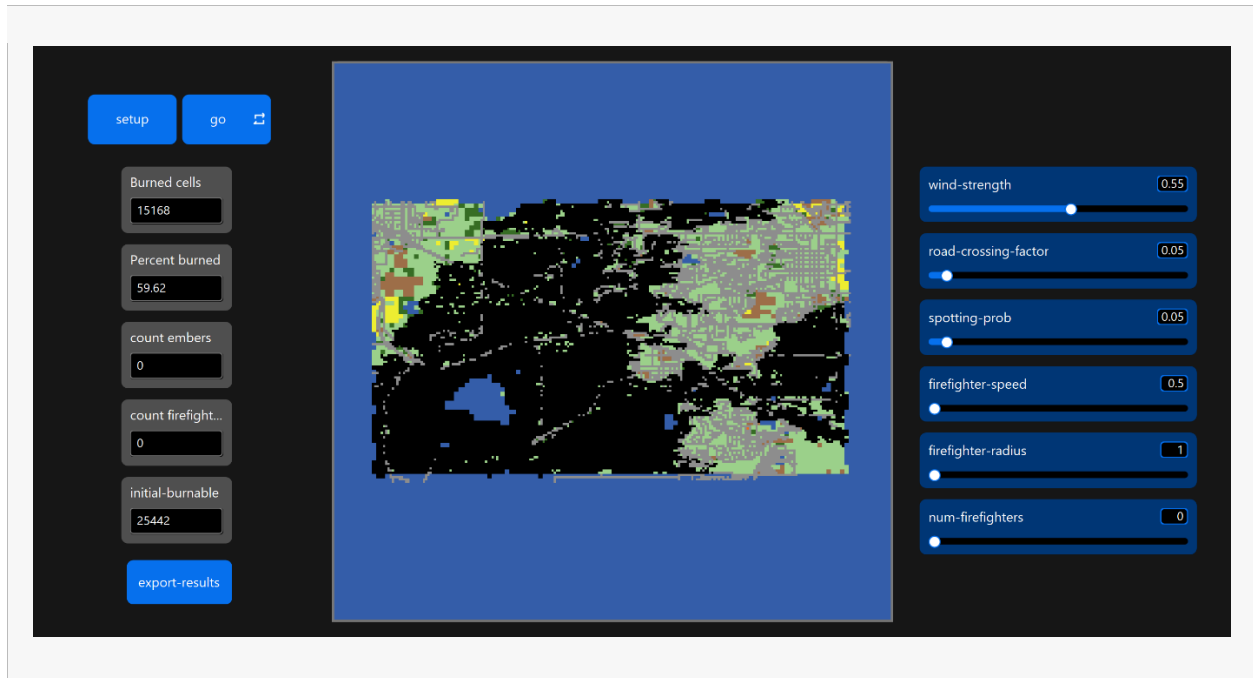


Figure 5. NetLogo model world at end of run with no firefighters. Black patches represent burned area. This run represents the baseline fire spread scenario.

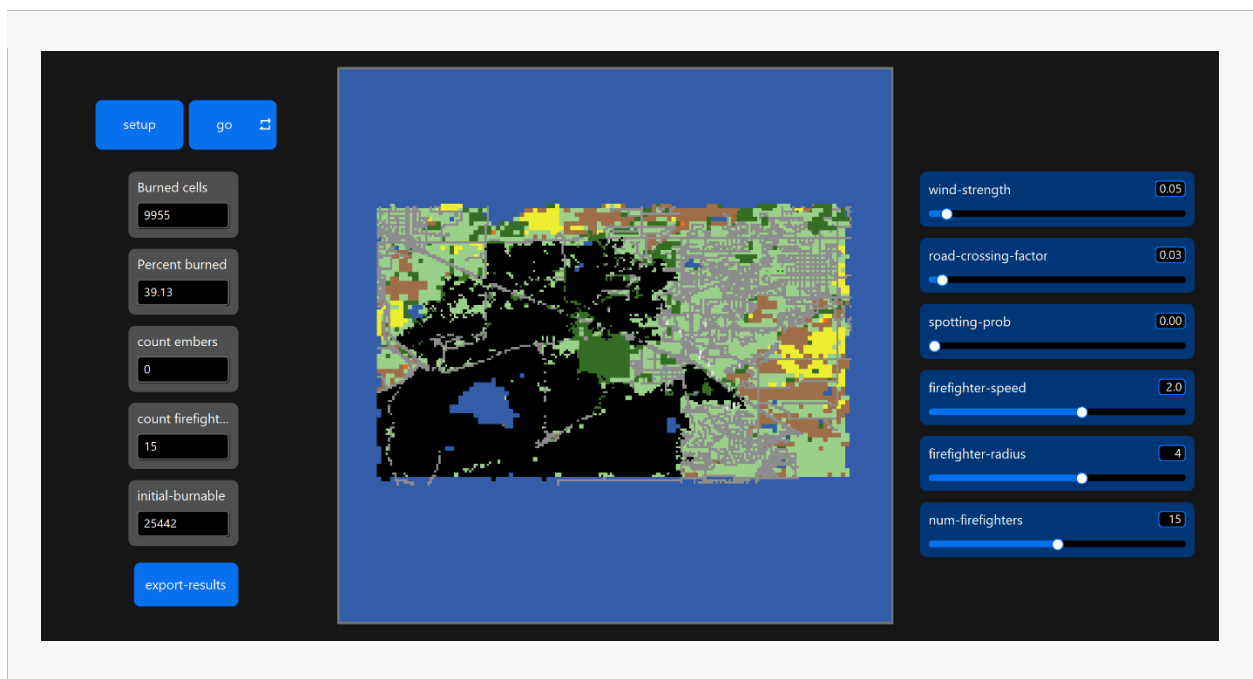


Figure 6. NetLogo model world at end of run with 15 firefighters active. Comparison with Figure 5 shows reduced burned area extent attributable to suppression activity.

4.3 Parameter Comparison

The following table summarizes results across model runs varying the number of firefighters while holding all other parameters constant.

Run	num-firefighters	wind-strength	burned-cells	% burned
1	0	0.0	4968	25.7
2	0	0.1	9858	38.75
3	15	0.05	9955	39.13
4	20	0.05	9010	35.4
5	5	0.15	10069	40

Table 2. Parameter comparison across model runs. Fill in burned-cells and % burned from your actual simulation results.

4.4 Output Map

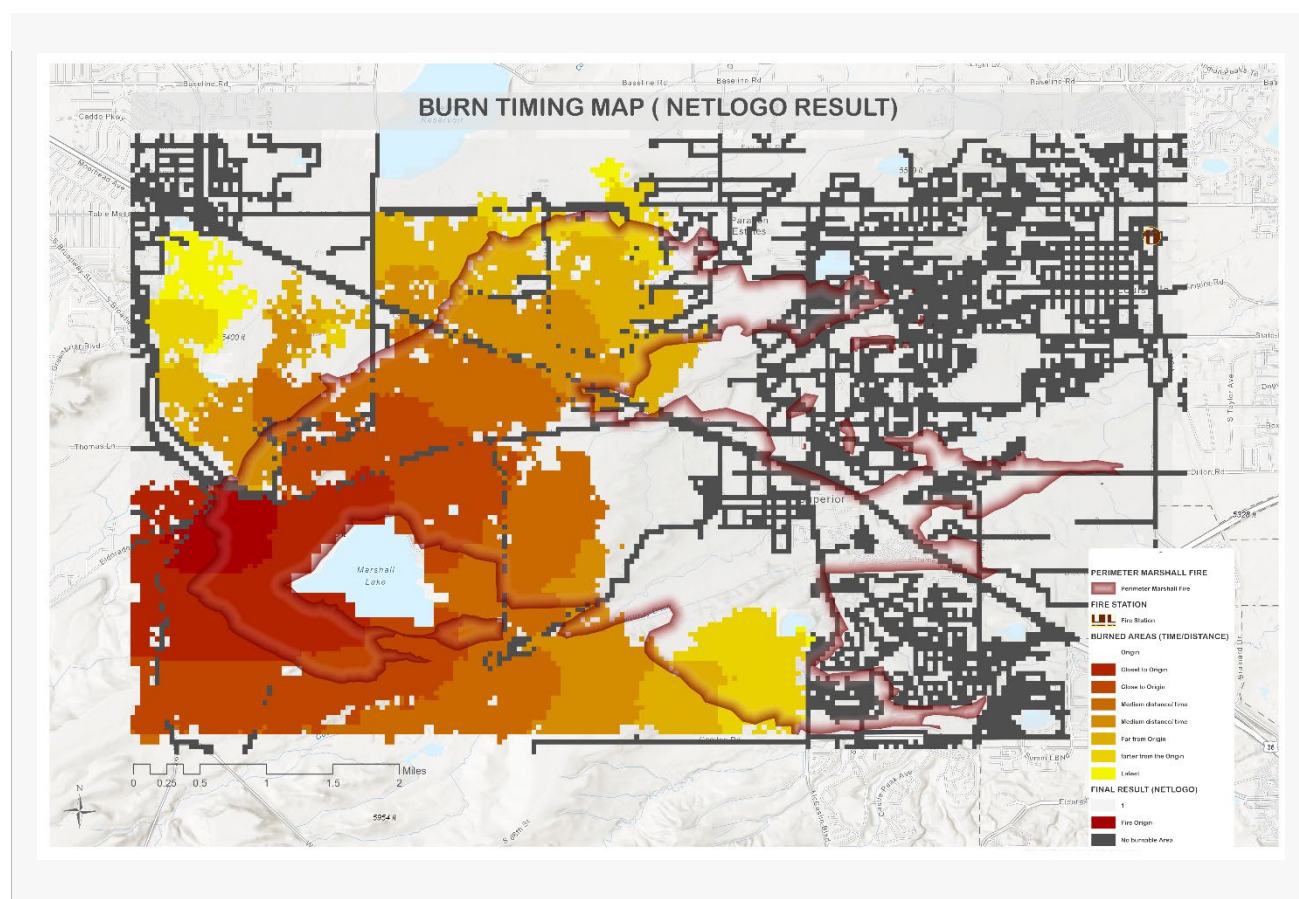


Figure 8. Direct comparison of modeled burn extent against the observed NIFC Marshall Fire perimeter. Areas of agreement and divergence between the simulated and documented fire are annotated.

5. Discussion

The model produced a spatially coherent fire spread pattern that reflected the main documented characteristics of the Marshall Fire at a qualitative level. Spread was predominantly eastward from the origin area, consistent with the strong westerly forcing described in National Weather Service summaries of the event (National Weather Service Boulder 2021). Grassland and shrubland patches supported rapid advance while developed patches showed fragmented or delayed ignition. Roads introduced visible directional effects on the fire perimeter shape, though these effects were partial rather than absolute as intended. The firefighter agents demonstrated meaningful suppression effects when present in sufficient numbers, with suppression becoming visible in the spatial pattern of the burned area at higher crew counts.

Several limitations of the model should be stated clearly. The flammability values assigned to NLCD classes are experimental rather than empirically calibrated fire behavior coefficients. They encode a plausible relative ordering of combustibility but do not correspond to standard fire behavior fuel models or site-specific measurements. The road representation is simplified: at the cell size used, road patches are wider than real roads, which exaggerates their influence on spread geometry. This is why road-crossing-factor was kept low in the final parameter settings, with spotting and wind doing more of the work in reproducing the observed pattern. The firefighter agents do not model water supply, engine capacity, tactical decisions, command structure, road closures, or building triage. They suppress nearby flames according to proximity rules, which is appropriate as an ABM interaction mechanism but is not a realistic emergency operations model.

Wind in the model is represented as a directional bias and spotting probability rather than as true atmospheric forcing. The National Weather Service documentation for the Marshall Fire describes a complex mountain-wave windstorm with spatially shifting wind fields and extreme gusts that are not captured by a constant directional vector (National Weather Service Boulder 2021). The ignition location is also approximate: the model places the origin near the Marshall area based on reported event geography, but the documented cause-and-origin investigation identified two separate ignition events that merged, a complexity that the model does not represent (Boulder County Sheriff's Office 2023).

Despite these limitations, the model achieves the core educational objectives of the assignment. It links real GIS data to a NetLogo simulation, assigns spatially differentiated behavior to patches from a land cover surface, introduces a second agent type with adaptive movement rules, and produces outputs that can be compared with observed data in a GIS environment. The Marshall Fire case study was particularly well suited to this exercise because its spread logic is strongly legible in spatial terms even at a simplified level. This legibility allowed a relatively simple ABM to generate interpretable results in relation to a real event, which is the central purpose of the spatial ABM approach as described by Crooks et al. (2018, 7).

Future improvements could include calibrated fuel models based on published fire behavior literature, finer-resolution spatial data to reduce geometric distortion of road patches, wind fields derived from meteorological records rather than a constant directional vector, and more realistic firefighter procedures incorporating route optimization, crew rotation, and resource constraints. Comparison of simulated burn timing against documented fire progression records would also provide a stronger quantitative validation of the spread dynamics. This type of model has clear real-world relevance for emergency planning, evacuation logistics, and infrastructure siting in communities at the wildland-urban interface, which is an increasingly consequential landscape type in the western United States (Natural Hazards Center 2022, 15).

References

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Appendix A: Interface Tab Configuration

Control	Type	Variable	Range / Default
Setup	Button	setup	Once
Go	Button	go	Forever
Export	Button	export-results	Once
num-firefighters	Slider	num-firefighters	0 to 30, default 5
wind-strength	Slider	wind-strength	0 to 1, default 0.4
road-crossing-factor	Slider	road-crossing-factor	0 to 1, default 0.15
spotting-prob	Slider	spotting-prob	0 to 0.5, default 0.08
firefighter-speed	Slider	firefighter-speed	0.5 to 5, default 2
firefighter-radius	Slider	firefighter-radius	1 to 8, default 3
Burned cells	Monitor	burned-cells	Display only
% burned	Monitor	(burned-cells / initial-burnable) * 100	Display only

Table A1. Interface tab controls and configuration for the Marshall Fire ABM model.

Appendix B: Final NetLogo Code

The complete final NetLogo code is reproduced below. All procedures are included.

```
extensions [gis]

globals [
  nlcd-dataset
  roads-dataset
  initial-burnable
  burned-cells
  ignition-lat
  ignition-lon
  wind-dir-x
  wind-dir-y
  station1-px
  station1-py
  station2-px
  station2-py
]

breed [fires fire]
breed [embers ember]
breed [firefighters firefighter]

firefighters-own [
  energy
  engaged?
  squad-id
  stuck-count
  last-target-distance
]

patches-own [
  lc
  road?
  base-flammability
  flammability
  burnable?
  final-state
  burn-tick
]

to setup
  clear-all
  set-default-shape turtles "square"

  gis:set-coordinate-system
    "GEOGCS[\"GCS_WGS_1984\",DATUM[\"D_WGS_1984\",SPHEROID[\"WGS_1984\",6378137,298.257223563]],PRIMEM[\"Greenwich\",0],UNIT[\"Degree\",0.0174532925199433]]"

  set nlcd-dataset gis:load-dataset "lc84.asc"
  set roads-dataset gis:load-dataset "roads_netlogo.shp"

  gis:set-world-envelope gis:envelope-of nlcd-dataset
  gis:apply-raster nlcd-dataset lc

  ;; approximate Marshall Fire origin
  set ignition-lat 39.952709
  set ignition-lon -105.231

  ;; default wind direction: east
  set wind-dir-x 1
  set wind-dir-y 0

  ask patches [
    set road? gis:intersects? roads-dataset self
    set base-flammability flammability-from-lc lc
    set flammability base-flammability

    ;; roads reduce spread but do not completely block it
    if road? [
      set flammability base-flammability * road-crossing-factor
    ]

    set burnable? (flammability > 0)
    set final-state -1
    set burn-tick -1
    recolor-patch
  ]

  set initial-burnable count patches with [burnable?]
  set burned-cells 0

  reset-ticks
  ignite-origin-cluster
  set-fire-station-locations

  if num-firefighters > 0 [
    let station1 patch station1-px station1-py
    let station2 patch station2-px station2-py

    create-firefighters num-firefighters [
      set color white
      set shape "person"
      set size 2.5
      set energy 300
      set engaged? false
      set squad-id floor (who / 5)
      set stuck-count 0
      set last-target-distance 999999

      ifelse (who mod 2 = 0)
      [ move-to station1 ]
      [ move-to station2 ]
    ]
  ]
end
```

```

]
]

update-final-state
end

to set-fire-station-locations
  let study-patches patches with [burnable? or road?]

  if any? study-patches [
    let max-x max [pxcor] of study-patches
    let max-y max [pycor] of study-patches
    let mid-x mean [pxcor] of study-patches
    let mid-y mean [pycor] of study-patches

    ;; station 1 = top-right of study area
    set station1-px (max-x - 3)
    set station1-py (max-y - 3)

    let candidate1 patch station1-px station1-py
    if candidate1 = nobody or not ([burnable? or road?] of candidate1) [
      let fallback1 min-one-of study-patches [
        distancexy station1-px station1-py
      ]
      if fallback1 != nobody [
        set station1-px [pxcor] of fallback1
        set station1-py [pycor] of fallback1
      ]
    ]

    ;; station 2 = right side, middle area
    let right-side-roads patches with [
      road? and
      pxcor > mid-x and
      pycor > (mid-y - 8) and
      pycor < (mid-y + 8)
    ]

    if any? right-side-roads [
      let s2 one-of right-side-roads
      set station2-px [pxcor] of s2
      set station2-py [pycor] of s2
    ]

    if not any? right-side-roads [
      let fallback2 patches with [road? and pxcor > mid-x]
      if any? fallback2 [
        let s2 min-one-of fallback2 [distancexy max-x mid-y]
        set station2-px [pxcor] of s2
        set station2-py [pycor] of s2
      ]
    ]

    if station2-px = nobody or station2-py = nobody [
      set station2-px station1-px
      set station2-py station1-py
    ]
  ]
end

to go
  if not any? fires and not any? embers [
    update-final-state
    stop
  ]

  move-firefighters

  if not any? fires and not any? embers [
    update-final-state
    stop
  ]

  ask fires [
    spread-fire
    set breed embers
  ]

  fade-embers
  update-final-state
  tick
end

to ignite-origin-cluster
  let ignition-xy gis:project-lat-lon ignition-lat ignition-lon

  if empty? ignition-xy [
    user-message "Ignition point is outside the model world."
    stop
  ]

  let center-patch patch (round item 0 ignition-xy) (round item 1 ignition-xy)

  if center-patch = nobody [
    user-message "Could not assign ignition point to a patch."
    stop
  ]

  ask patches with [
    distance center-patch <= 2 and
    burnable? and
    flammability >= 0.35
  ] [
    ignite
  ]
end

to spread-fire
  let source-patch patch-here

```

```

ask neighbors with [burnable? and pcolor != black and not any? turtles-here] [
  let step-x (pxcor - [pxcor] of source-patch)
  let step-y (pycor - [pycor] of source-patch)

  let directional-bonus 0

  if (step-x = wind-dir-x and step-y = wind-dir-y) [
    set directional-bonus wind-strength
  ]

  if ((step-x = wind-dir-x and step-y = 0) or
      (step-x = 0 and step-y = wind-dir-y)) [
    set directional-bonus max list directional-bonus (wind-strength * 0.55)
  ]

  ;; slowed down more so crews can matter
  let spread-prob min list 1 (flammability + directional-bonus + 0.02)

  if random-float 1 < spread-prob [
    ignite
  ]
]

;; spotting allows jumps across barriers
if wind-strength > 0 [
  let spot-distance (2 + random 3)
  let side-drift one-of [-1 0 1]

  let spot-patch patch-at
    (wind-dir-x * spot-distance + ifelse-value (wind-dir-y = 0) [0] [side-drift])
    (wind-dir-y * spot-distance + ifelse-value (wind-dir-x = 0) [0] [side-drift])

  if spot-patch != nobody [
    ask spot-patch [
      if burnable? and pcolor != black and not any? turtles-here [
        if random-float 1 < spotting-prob [
          ignite
        ]
      ]
    ]
  ]
]

end

to ignite
  if not burnable? [ stop ]
  if pcolor = black [ stop ]

  if burn-tick = -1 [
    set burn-tick ticks
  ]

  sprout-fires 1 [
    set color red
    set size 3
    set shape "square"
  ]

  color-burned-patch
  set burned-cells burned-cells + 1
end

to color-burned-patch
  set pcolor black
end

to fade-embers
  ask embers [
    set color color - 0.25
    if color < (red - 3.5) [
      die
    ]
  ]
end

to move-firefighters
  if not any? firefighters [ stop ]
  if not any? fires and not any? embers [ stop ]

  ask firefighters [
    let target nobody
    let my-squad squad-id
    let active-targets (turtle-set fires embers)

    if any? active-targets [
      let target-options nobody

      if my-squad mod 3 = 0 [
        set target-options active-targets with [xcor <= mean [xcor] of active-targets]
      ]
      if my-squad mod 3 = 1 [
        set target-options active-targets with [ycor >= mean [ycor] of active-targets]
      ]
      if my-squad mod 3 = 2 [
        set target-options active-targets with [ycor < mean [ycor] of active-targets]
      ]

      if any? target-options [
        set target min-one-of target-options [distance myself]
      ]
      if target = nobody [
        set target min-one-of active-targets [distance myself]
      ]
    ]

    if target != nobody [
      let current-dist distance target

      if current-dist >= last-target-distance - 0.1 [
        set stuck-count stuck-count + 1
      ]
    ]
  ]
end

```

```

]
if current-dist < last-target-distance - 0.1 [
  set stuck-count 0
]
set last-target-distance current-dist

if distance target <= firefighter-radius + 1 [
  set engaged? true
]

ifelse engaged? [
  ;; once engaged, move freely near the fire
  face target
  fd firefighter-speed
] [
  ;; before engagement, prefer roads but also allow open movement
  let travel-options patches in-radius firefighter-speed with [
    road? or
    lc = 21 or
    lc = 22 or
    lc = 23 or
    lc = 31 or
    lc = 71 or
    lc = 81
  ]

  if any? travel-options [
    let better-options travel-options with [distance target < current-dist]

    if any? better-options [
      let next-step one-of better-options
      move-to next-step
    ]
    if not any? better-options [
      let next-step min-one-of travel-options [distance target]
      move-to next-step
    ]
  ]

  if stuck-count >= 4 [
    face target
    fd firefighter-speed
    set stuck-count 0
    set engaged? true
  ]
]

ask fires in-radius firefighter-radius [
  die
]

ask embers in-radius firefighter-radius [
  die
]

ask patches in-radius firefighter-radius with [burnable? and pcolor != black] [
  set flammability max list 0 (base-flammability * 0.35)
  set burnable? (flammability > 0)
  recolor-patch
]

set energy energy - 1
if energy <= 0 [
  die
]
]
end

to update-final-state
  ask patches [
    if not burnable? [
      set final-state 0
    ]
    if burnable? [
      set final-state 1
    ]
    if road? [
      set final-state 3
    ]
    if pcolor = black [
      set final-state 2
    ]
  ]
end

to export-results
  update-final-state
  gis:store-dataset (gis:patch-dataset final-state) "final_output1.asc"
  gis:store-dataset (gis:patch-dataset burn-tick) "burn_tick1.asc"
end

to recolor-patch
  if road? and flammability <= 0.05 [
    set pcolor gray
    stop
  ]

  if not burnable? [
    set pcolor blue
    stop
  ]

  if flammability < 0.15 [
    set pcolor 57
    stop
  ]

  if flammability < 0.35 [
    set pcolor 53
  ]

```

```
    stop
  ]

  if flammability < 0.60 [
    set pcolor 45
    stop
  ]

  if flammability < 0.80 [
    set pcolor 35
    stop
  ]

  set pcolor 25
end

to-report flammability-from-lc [v]
  if v = 11 [ report 0 ]

  if v = 21 [ report 0.12 ]
  if v = 22 [ report 0.10 ]
  if v = 23 [ report 0.07 ]
  if v = 24 [ report 0.04 ]

  if v = 31 [ report 0.12 ]

  if v = 41 [ report 0.78 ]
  if v = 42 [ report 0.88 ]
  if v = 43 [ report 0.82 ]

  if v = 52 [ report 0.70 ]
  if v = 71 [ report 0.68 ]
  if v = 81 [ report 0.48 ]
  if v = 82 [ report 0.38 ]

  if v = 90 [ report 0.22 ]
  if v = 95 [ report 0.12 ]

  report 0
end
```